

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Summer Examination – 2023 Course: B. Tech. Branch : Computer Science & Engineering Semester :VI Subject Code & Name: BTCOE604B - Internet of Things Max Marks: 60 Date: /07/2023 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Differentiate between IT and OT networks.	(Understand)	6
B)	Draw and explain core IoT functional Stack.	(Understand)	6
C)	Draw & explain IoT World Forum (IoTWF) Standardized Architecture.	(Understand)	6
Q.2	Solve Any Two of the following.		12
A)	Explain different types of sensors.	(Remember)	6
B)	Explain components of smart objects.	(Understand)	6
C)	Describe various Communications Criteria.	(Understand)	6
Q. 3	Solve Any Two of the following.		12
A)	Explain Key Advantages of Internet Protocol.	(Remember)	6
B)	Discuss Message Queuing Telemetry Transport (MQTT).	(Understand)	6
C)	Illustrate any two techniques of IP Optimization for IoT.	(Application)	6
Q.4	Solve Any Two of the following.		12
A)	What is HDFS in Hadoop? Explain namenode and datanode.	(Understand)	6
B)	What is Machine Learning? Explain the role of Machine Learning in IoT.	(Understand)	6
C)	What is Edge analytics? Describe Edge Analytics Core Functions.	(Understand)	6
Q. 5	Solve Any Two of the following.		12
A)	Illustrate basic structure of Arduino programming with LED blink program?	(Application)	6
B)	What is Raspberry pi? Draw hardware layout of raspberry pi.	(Remember)	6
C)	What is Temperature Sensor? Draw interfacing of any one temperature sensor with Arduino with code.	(Remember)	6